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BEE SWARMING AND ABSCONDMENT DETECTION SYSTEM (BSADS)

SOFTWARE DESIGN DOCUMENT (SDD)

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1. INTRODUCTION

1.1 Purpose of the Document

The purpose of this Software Design Document (SDD) is to describe the architectural and detailed design of the Bee Swarming and Abscondment Detection System. This document translates the requirements specified in the Software Requirements Specification (SRS) into a technical design that will guide system implementation.

1.2 Scope of the System

The **Bee Swarming and Abscondment Detection System** (BSADS) is an Internet of Things (IoT)–based monitoring solution designed to detect early signs of bee swarming and abscondment in beehives. The system shall integrate environmental and acoustic sensors, a machine-learning-based analysis module, and a mobile application to provide real-time insights and alerts to beekeepers.

The operational scope of the system shall begin at the point where raw hive data (acoustic recordings and associated environmental measurements) are retrieved from their storage location, whether collected through deployed sensors or obtained from previously stored datasets. However, not all audio recordings shall be guaranteed to have corresponding environmental parameters. Therefore, the system shall be designed to process acoustic data independently, while incorporating environmental variables only when such data are available.

Once retrieved, the raw data undergoes preprocessing, feature extraction, and analytical processing within a cloud-based or centralized computing platform. The processed data are then subjected to machine-learning–based classification to determine the hive state as one of the following categories: Normal, Pre-swarm, Swarm, Abscondment, or Uncertain. The system further supports real-time data visualization, predictive alert notifications, and historical analysis through a user-friendly mobile application.

The primary goal of the project is to reduce colony loss by enabling early detection of abnormal hive behavior and supporting timely beekeeper intervention. The objectives of the system include

improving the accuracy of hive monitoring, minimizing the need for frequent manual inspections, and supporting data-driven decision-making in apiculture.

The key benefits of the system include enhanced colony retention, improved operational efficiency for beekeepers, contribution to apicultural research through long-term data collection, and promotion of smart and sustainable agricultural practices.

1.3 Overview of the Document

This Software Design Document (SDD) provides a detailed description of the design and architecture of the Bee Swarming and Abscondment Detection System. The document translates the functional and non-functional requirements defined in the Software Requirements Specification (SRS) into a structured technical design that guides system implementation.

This Software Design Document (SDD) is organized into seven structured chapters, each addressing a specific aspect of the BSADS design.

i) **Chapter 1: Introduction** presents the purpose, scope, reference materials, and definitions relevant to the system. It establishes the context of the document and clarifies the terminology and standards adopted throughout the design process.

ii) **Chapter 2: System Overview** provides a high-level description of the BSADS system, including the system context diagram, product functions, user classes and characteristics, operating environment, and design and implementation constraints. This chapter describes the overall functionality and positioning of the system before detailing its internal structure.

iii) **Chapter 3: System Architecture** describes the architectural design of the system. It includes the architectural model, subsystem decomposition, data flow diagrams, structural decomposition, and the rationale behind architectural decisions. Trade-offs and alternative architectural approaches considered during the design phase are also discussed.

iv) **Chapter 4: Data Design** defines how system data is structured and managed. It includes detailed descriptions of data entities, data organization, and the system data

dictionary, specifying attributes and relationships among core entities such as users, sensor data, inference data, and alerts.

v) **Chapter 5: Component Design** provides a detailed description of individual system components, including data integration, preprocessing and feature extraction, machine learning inference, data storage, cloud synchronization, backend services, mobile application modules, and system monitoring. Each component's purpose, inputs, outputs, responsibilities, and processing logic are defined.

vi) **Chapter 6: Human Interface Design** describes the user interface structure of the system, including authentication screens, monitoring dashboards, alert displays, and configuration interfaces. This chapter defines screen objects, user actions, and interaction flow between users and the system.

vii) **Chapter 7: Requirements Matrix:** Provides a traceability matrix to ensure all functional requirements from the SRS are addressed within this design.

Together, these chapters provide a comprehensive technical blueprint that supports system development, testing, deployment, and maintenance.

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1.5 Definitions and Acronyms

Table 1.5.1 Definitions and acronyms

Term	Definition
Abscondment	The sudden abandonment of a beehive by the entire bee colony
API	Application Programming Interface
BSADS	Bee Swarming and Abscondment Detection System
DFD	Data Flow Diagram
Hive Node	Embedded hardware unit consisting of sensors and a microcontroller deployed in a beehive
IoT	Internet of Things
MFCC	Mel-Frequency Cepstral Coefficients

ML	Machine Learning
SDD	Software Design Document
SRS	Software Requirements Specification
Swarming	The natural process where a colony splits and forms a new colony

2. SYSTEM OVERVIEW

2.1 Problem Statement and System Description

2.1.1. Problem Statement

Beekeepers often lose colonies due to swarming and abscondment that are detected too late through manual inspections. Existing hive monitoring solutions collect acoustic and environmental data but lack intelligent audio data analytical capabilities capable of transforming raw hive data into early-warning insights for proactive decision-making.

The Bee Swarming and Abscondment Detection System (BSADS) addresses this gap by transforming stored hive data into intelligent, early-warning analytics that support timely beekeeper intervention and reduce colony loss.

2.1.2 System Description

BSADS is an Internet of Things (IoT) enabled analytical and decision-support system designed to assist beekeepers in detecting early signs of swarming and abscondment. The system operates on already collected hive datasets obtained from existing monitoring platforms or stored repositories. These datasets primarily include acoustic recordings and may, in some instances, be accompanied by environmental parameters such as temperature and relative humidity. Where environmental data is unavailable, the system performs acoustic-only analysis on the ingested audio data.

The objective of BSADS is not sensor deployment, but intelligent data interpretation. BSADS operates strictly as a data-driven analytical layer and does not directly interface with physical sensing hardware. The system operates through a set of functional modules organized as a sequential data-processing pipeline, where each module transforms inputs into progressively higher-level analytical outputs:

i) **Data ingestion & preprocessing module:** Responsible for collecting datasets from existing monitoring platforms and performing noise reduction, normalization, filtering, and segmentation to prepare data for analysis.

ii) **Data processing & machine learning module:** Performs feature extraction (e.g., dominant frequencies, spectral energy, temperature–humidity correlations) and applies trained machine learning models to classify hive conditions into operational states such as Normal, Pre-swarm, Swarm, Abscondment, or Uncertain.

iii) **Visualization and notification module**: Presents hive status through dashboards and generates alerts when abnormal conditions are detected.

iv) **Advisory module:** Provides actionable recommendations based on detected hive conditions and historical behavioral patterns.

v) **Geospatial mapping module:** Displays geographic locations of monitored hives on a map and their respective operational status indicators.

vi) **Connectivity & Synchronization module:** Enables temporary local caching of results during network interruptions (when the system goes offline) and automatic synchronization once connectivity is restored.

2.2 BSADS Context Diagram

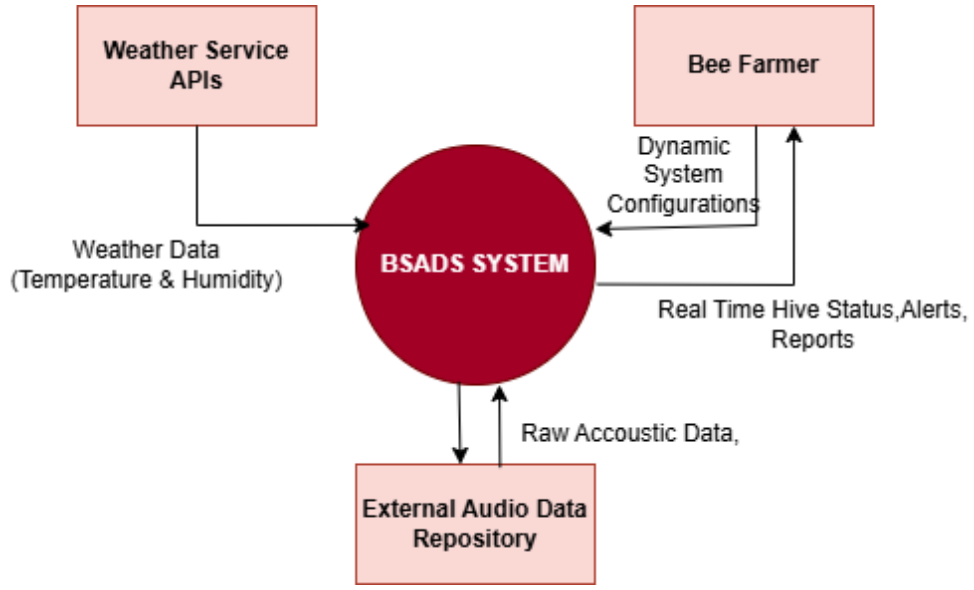

Figure 2. 1 System Context Diagram of the Bee Swarming and Abscondment Detection System

Figure 2.1 illustrates the overall operational context of the Bee Swarming and Abscondment Detection System (BSADS). The diagram presents the system as a single central process and shows its interaction with external entities through defined data flows.

The external entities include:

i) **Beehive Sensor Database**: A centralized data repository responsible for storing high-resolution raw acoustic signals captured from the hive, as well as associated environmental parameters such as temperature and humidity, where available. This dataset serves as the primary input for signal processing and machine learning models.

ii) **Weather Service APIs:** External data integration services that provide real-time and historical meteorological data (e.g., temperature, humidity, rainfall, and wind patterns). These inputs enhance contextual analysis and improve the robustness and accuracy of predictive models.

iii) **Bee Farmer (End User)**: The primary stakeholder interacting with the system, responsible for monitoring hive conditions. The system delivers actionable insights through alerts, analytical reports, and real-time status updates to support informed decision-making.

The diagram demonstrates how acoustic and environmental data flow from the bee hive sensor storage databases into the BSADS for preprocessing and analytical evaluation. External weather data is also integrated into the system to enhance contextual understanding of hive conditions. After preprocessing through noise reduction and feature extraction, machine learning models analyze the processed data to detect abnormal patterns and classify hive states. The resulting real-time status updates, predictive alerts, and analytical reports are transmitted to the Bee Farmer through the mobile application interface.

The diagram further illustrates bidirectional communication, where the Bee Farmer may send alert acknowledgments or configuration settings to the system. Additionally, the system does not directly control or interact with physical hive sensors but relies entirely on externally provided datasets.

2.3 Product Functions

The functional modules described in this section expand upon the high-level pipeline introduced in *Section 2.1.2* and provide detailed responsibilities for each stage of the system.

The BSADS architecture is organized into distinct functional modules, each responsible for a specific stage of the data processing and decision-support pipeline. These modules operate in coordination to transform raw hive datasets into meaningful analytical outputs, visual insights, and actionable recommendations for beekeepers.

Specifically, the system comprises the following functional modules:

i) Data ingestion & preprocessing

This module is responsible for acquiring raw datasets from existing hive monitoring platforms and preparing them for analytical processing. Since raw data may contain inconsistencies, noise, or incomplete records, preprocessing is necessary to ensure reliability and analytical accuracy.

Functions include:

a) Retrieval of stored acoustic and environmental datasets
b) Validation of data completeness, structure, and integrity
c) Noise filtering and removal of corrupted or duplicate entries
d) Normalization and standardization of data formats
e) Segmentation of data into analyzable intervals

ii) Data preprocessing and machine learning

This module performs computational analysis on the processed datasets to detect behavioral patterns associated with swarming and abscondment. It extracts relevant features and applies trained machine learning models to determine the operational state of the hive.

Functions include:

1. Extraction of acoustic frequency features (e.g., dominant frequencies, band energy, spectral centroid, spectral flux variations, temporal characteristics)
2. Correlation of available environmental parameters (e.g., temperature and humidity)
3. Classification of hive states into:

    - Normal
    - Pre-swarm
    - Swarm
    - Abscondment
    - Uncertain

iii) Visualization and Notification

This module translates analytical outputs into user-friendly representations accessible through the mobile application. It ensures that complex analytical results are presented in an interpretable and actionable format.

Functions include:

- Presentation of hive condition analytics via dashboards
- Display of behavioral trends and risk indicators
- Generation of predictive alerts upon detection of abnormal patterns

iv) Advisory Support

This module provides decision-support guidance based on the classified hive state. It assists beekeepers in taking appropriate and timely action to prevent colony loss.

Functions include:

- Provision of recommended interventions based on detected hive conditions
- Guidance on preventive measures and best management practices

v) Geospatial Mapping

This module provides spatial awareness of hive conditions by linking analytical results to geographic locations.

Functions include:

- Visualization of hive locations on a digital map
- Spatial representation of colony risk levels and status indicators

vi) Connectivity & Synchronization

This module ensures continuity of system access during temporary network interruptions.

Functions include:

- Local access to previously synchronized analytical results and alerts
- Storage of updates during offline operation
- Automatic synchronization with the central platform once connectivity is restored

Additionally, historical data and classification results are stored to support long-term performance evaluation and research.

2.4 User Classes and Characteristics

The BSADS is intended for multiple user categories, each with distinct roles, responsibilities, and interaction requirements. The system design accommodates varying levels of technical expertise to ensure usability and effectiveness across all user groups.

2.4.1. Beekeepers (End Users)

Beekeepers are the primary users of the system. They interact with the BSADS through the mobile application to monitor hive conditions, receive alerts, and review historical data. Beekeepers may possess limited technical knowledge; therefore, the user interface is designed to be intuitive, visually informative, and easy to navigate. The system provides clear notifications and simplified

data visualizations to support timely decision-making without requiring advanced technical skills. These also administer the system through configuring the system, managing user access and settings, among others.

2.4.2. Researchers and Analysts

Researchers and analysts use the system to access aggregated and historical data for research, evaluation, and model improvement purposes. This user class typically has advanced analytical skills and may interact with exported datasets or analytical dashboards. The system supports this group by maintaining structured data storage and providing access to long-term records necessary for evaluating hive behavior and machine learning model performance.

Table 2. 1 Users of the BSADS system and their technical expertise level

User Class	Primary Interaction Method	Technical Expertise Level
Beekeeper	Mobile Application	Low to Medium
Researcher	Data Analytics and Reporting Tools	High

2.4 Operating Environment

The BSADS operates within a cloud-centered software environment that supports data storage, machine learning analytics, visualization services, and mobile access. The operating environment consists of the following components:

a) Cloud Data Storage Environment
b) Machine Learning Processing Environment
c) Visualization and Backend Support Environment
d) Mobile Application Environment
e) Communication Infrastructure

Each environment above is as described below.

2.4.1. Cloud Data Storage Environment

The system operates on pre-collected acoustic datasets, some of which may optionally include corresponding environmental parameters such as temperature and humidity. These datasets are stored in a cloud-based storage infrastructure. The cloud storage environment provides:

- Centralized storage of acoustic and environmental datasets
- Scalable database services for structured and unstructured data
- Secure data access control mechanisms
- Long-term archival for historical analysis

This environment forms the starting point of the BSADS data pipeline, as the system retrieves raw datasets directly from this storage layer.

2.4.2. Machine Learning Processing Environment

The analytics component operates within a cloud-based computational environment that supports preprocessing and machine learning inference. This environment provides:

- Data preprocessing capabilities (cleaning, normalization, feature extraction)
- Execution of trained machine learning classification models
- Computational resources for batch or near-real-time analysis
- Model storage and version management

The machine learning environment is responsible for transforming raw hive datasets into classified operational states (Normal, Pre-swarm, Swarm, Abscondment, or Uncertain).

2.4.3 Visualization and Backend Support Environment

The system includes backend services that act as an intermediary between the analytics engine and the mobile application interface. This environment provides:

- Application Programming Interfaces (APIs) for data access
- Business logic for alert generation

- Dashboard data preparation
- Secure authentication and authorization services

It ensures that analytical results are structured and delivered efficiently to end users.

2.4.4 Mobile Application Environment

The mobile application operates on Android-based smartphones and serves as the primary user interface for beekeepers and system stakeholders. The mobile environment supports:

- Dashboard visualization of hive status
- Alert notifications and advisory messages
- Access to historical trends and reports
- Secure user authentication

Internet connectivity is required for real-time data synchronization, although limited offline access may be supported for previously retrieved results.

2.4.5 Communication Infrastructure

The system relies on internet-based communication networks to enable interaction between cloud services and client applications. The communication infrastructure supports:

- Secure data transfer using encrypted protocols
- API-based communication between backend services and mobile devices
- Synchronization of analytical results and alerts

The design accounts for intermittent connectivity by allowing delayed synchronization when network access becomes available.

2.5 Design and Implementation Constraints

The design and implementation of the BSADS are subject to several technical, operational, and project-related constraints that influence architectural decisions, system behavior, and achievable performance. These constraints are enumerated as follows:

i) **Computational and Storage Constraints**:

The BSADS relies on cloud-based processing for data analysis and machine-learning inference due to the limited processing power, memory, and storage available on end-user mobile devices. Smartphones are therefore used primarily for visualization, alert delivery, and user interaction rather than for executing computationally intensive analytics or machine learning models. To address these limitations, trained models are deployed within the cloud environment, and only summarized results, alerts, and selected historical data are synchronized to the mobile application to minimize storage and processing overhead.

ii) **Communication and Connectivity Constraints**:

The system depends on internet connectivity to enable communication between cloud services and the mobile application. In rural or remote apiary locations, network instability or limited connectivity may affect real-time access to analytics and alerts. To mitigate this constraint, the system supports intermittent connectivity by allowing delayed synchronization and limited offline access to previously retrieved results, with automatic updates once network access is restored.

iii) **Data Quality and Model Accuracy Constraints:**

The accuracy of the machine learning models is influenced by the quality, completeness, and representativeness of the available datasets. Some acoustic datasets may lack corresponding environmental parameters, and certain hive states, such as abscondment, may be underrepresented. These factors can limit classification confidence and predictive accuracy. As a result, the system incorporates uncertainty handling in classification outputs

and is designed to support future model retraining as additional and higher-quality data become available.

iv) **Advisory Scope Constraints:**

The advisory component provides guidance based on detected hive states and predefined knowledge rules. However, it cannot account for all possible biological, environmental, or management scenarios encountered in real-world beekeeping practices. The advisory outputs are therefore intended to support informed decision-making rather than replace expert judgment or beekeeper experience.

v) **Project and Resource Constraints**:

Project-related limitations, including development time, budget, availability of labeled datasets, and deployment resources, influence system scalability, model complexity, and the extent of real-world validation during the implementation phase.

The functional decomposition presented in this chapter forms the foundation for the architectural design described in *Chapter 3*, where each functional module is mapped to a corresponding subsystem to ensure consistency between system functionality and structural implementation.

3. SYSTEM ARCHITECTURE

3.1 Architectural Design

The BSADS is designed using a modular, layered architecture in which system responsibilities are partitioned into well-defined and cooperating subsystems. This architectural approach is guided by the need to separate concerns related to data integration, analytical processing, persistent storage, system coordination, and user interaction. Such separation enhances system maintainability, scalability, fault isolation, and supports independent evolution of system components over time.

The architectural design presented in this section is derived directly from the functional decomposition defined in Chapter 2. Each subsystem corresponds to one or more functional modules, thereby ensuring consistency and traceability between system functionality and structural implementation.

At a high level, BSADS is decomposed into six major subsystems:

1. External Hive Monitoring Interface
2. Data Integration and Communication Subsystem
3. Backend Coordination Subsystem
4. Data Processing and Machine Learning Subsystem
5. Cloud Data Management Subsystem
6. Application Presentation Subsystem

Each subsystem is assigned clearly defined responsibilities and interacts with other subsystems through well-defined interfaces. This structured decomposition ensures that data flows in a controlled and predictable manner across the system while maintaining clear boundaries between responsibilities.

3.1.1 Functional to Architectural Mapping

To ensure consistency between the system's functional design (Chapter 2) and its architectural structure, each functional module is mapped to a corresponding subsystem. This mapping demonstrates how high-level system functionalities are realized within the architectural components.

Table 3. 1: Functional modules to architectural subsystem mapping

Functional Module (Section 2.3)	**Corresponding Subsystem (Chapter 3)**
Data Integration & Preprocessing	Data Integration and Communication Subsystem
Data Processing & Machine Learning	Data Processing and Machine Learning Subsystem
Visualization & Notification	Application Presentation Subsystem and Backend Coordination Subsystem
Advisory Support	Backend Coordination Subsystem
Geospatial Mapping	Application Presentation Subsystem
Connectivity & Synchronization	Backend Coordination Subsystem and Application Presentation Subsystem
Data Storage (implicit)	Cloud Data Management Subsystem

Table 3.1 presents the mapping between the functional modules defined in Chapter 2 and the corresponding architectural subsystems in Chapter 3, illustrating how system functionalities are realized within the structural design of BSADS.

This mapping ensures that all functional requirements identified in Chapter 2 are fully represented within the system architecture, thereby maintaining structural and behavioral consistency across the system design.

3.1.2 Subsystem Responsibilities

i) External hive monitoring interface

The External Hive Monitoring Interface represents the boundary between BSADS and external hive monitoring platforms. It provides access to pre-collected acoustic datasets and, where available, associated environmental parameters such as temperature and humidity. This subsystem operates outside the direct control of BSADS and serves strictly as the source of input data. The system does not directly interact with or control physical sensing hardware but relies entirely on externally generated datasets.

ii) Data ingestion & communication subsystem

This acts as the controlled entry point into the system. It is responsible for receiving datasets from external monitoring platforms and ensuring that incoming data is valid, secure, and properly structured before further processing. Its primary responsibilities include:

- Receiving datasets from external monitoring systems
- Validating dataset structure and metadata
- Enforcing authentication and access control
- Assigning unique identifiers to incoming records
- Forwarding validated datasets to backend services

iii) Backend coordination subsystem

This serves as the central orchestrator of system operations. It manages workflow execution and coordinates interactions between different subsystems, ensuring that data flows correctly through the processing pipeline. Its responsibilities include:

- Routing validated datasets to preprocessing pipelines
- Triggering machine learning inference
- Managing alert-generation logic
- Providing RESTful APIs for client communication
- Coordinating interactions between analytics and storage layers

This subsystem acts as the central controller of the system.

iv) Data processing & machine learning subsystem

This forms the analytical core of BSADS. It is responsible for transforming raw acoustic datasets into meaningful insights through a series of computational processes.

This subsystem performs preprocessing operations such as noise reduction and signal normalization, extracts relevant acoustic features, integrates environmental parameters where available, and applies trained machine learning models to classify hive conditions. The output consists of hive-state classifications (e.g., Normal, Pre-swarm, Swarm, Abscondment, or Uncertain) along with associated confidence scores. Through these processes, the subsystem converts raw hive recordings into structured operational intelligence.

v) Cloud data management subsystem

This subsystem provides persistent storage and data organization capabilities for the system. It maintains all critical data required for both real-time operations and long-term analysis. It maintains:

- User and hive records
- Raw dataset metadata
- Extracted feature vectors
- Inference results
- Alert records and logs

It supports both real-time retrieval and long-term historical analysis.

vi) Application presentation subsystem

The application presentation subsystem represents the user-facing layer of the system, primarily implemented through a mobile application. It is responsible for presenting analytical outputs in an accessible and user-friendly format. It is responsible for:

- Displaying hive state classifications
- Presenting dashboards and historical trends
- Delivering alerts and advisory messages
- Supporting secure user authentication

It interacts exclusively with backend APIs and does not perform analytical processing locally.

3.1.3 Subsystem collaboration

The subsystems within BSADS operate in a coordinated and sequential manner to form a complete data processing and decision-support pipeline. The interaction begins with the *External Hive Monitoring Interface* providing raw datasets to the system. These datasets are received and validated by the *data integration and communication subsystem* before being forwarded to the Backend Coordination Subsystem.

The *backend coordination subsystem* then triggers preprocessing and analytical operations within the *data processing and machine learning subsystem*. Once analysis is complete, the resulting classifications and associated data are stored within the *cloud data management subsystem*.

Finally, the *application presentation subsystem* retrieves the processed results through backend APIs and presents them to the end user in the form of dashboards, alerts, and advisory messages. Throughout this process, the *backend coordination subsystem* acts as the central mediator, ensuring controlled and consistent data flow between all subsystems.

This coordinated interaction ensures that BSADS operates as an integrated system capable of transforming raw hive datasets into actionable insights for beekeepers while maintaining architectural consistency and data integrity.

3.1.4 BSADS High-Level Architectural Diagram

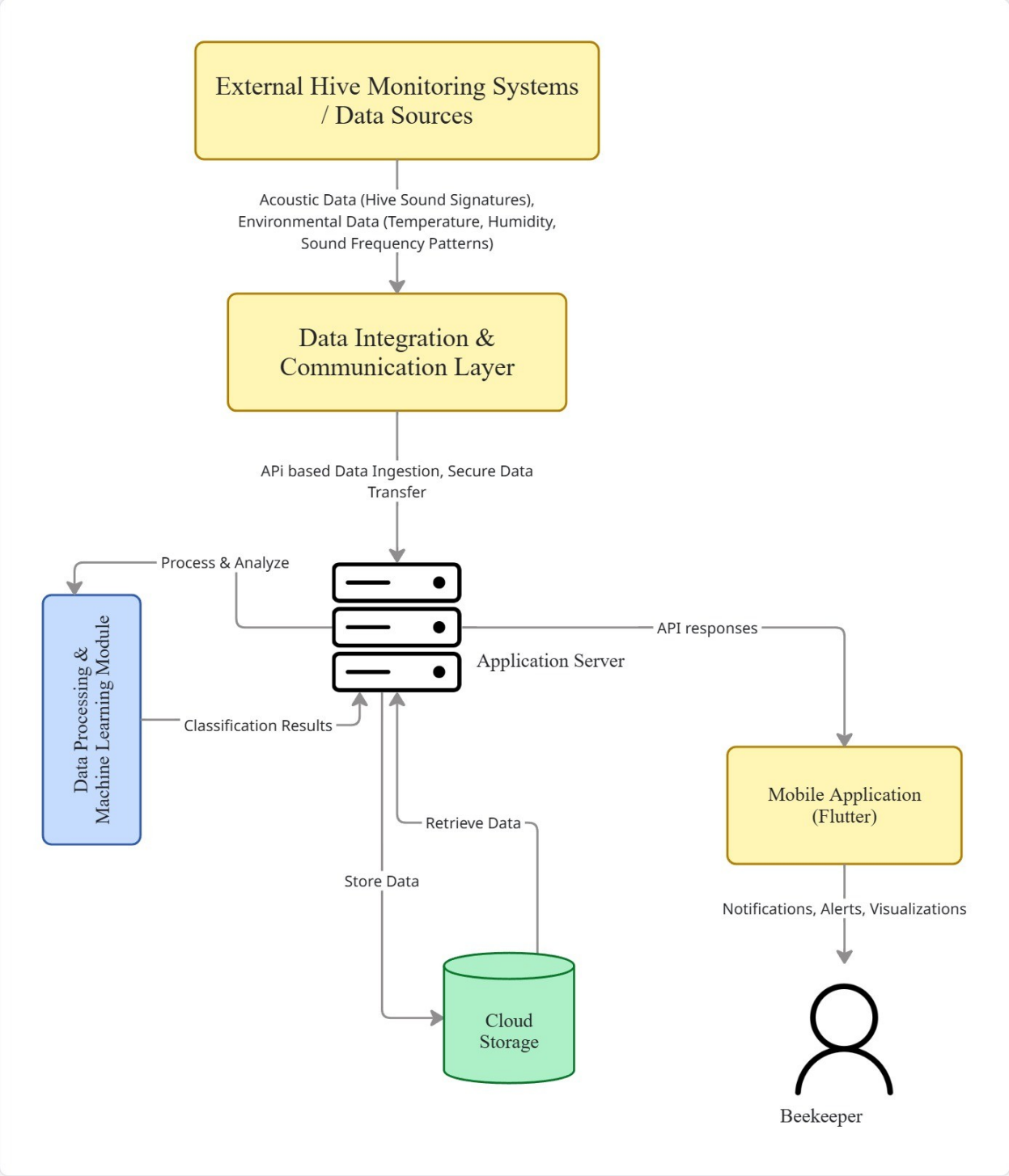

Figure 3. 1 High-Level Architectural Diagram for BSADS

Figure 3.1 illustrates the high-level architecture of the BSADS, showing the interaction between the external hive monitoring node or data source, communication layer, backend services, machine learning module, database, and mobile application.

3.2 Decomposition Description

This section presents a functional decomposition of the BSADS. The decomposition refines the high-level architectural structure defined in Section 3.1 into logically cohesive processing units and associated data repositories. It provides a detailed view of how data flows through the system and how individual processing stages collaborate to produce analytical outputs.

A functional modeling approach is adopted because the behavior of BSADS is primarily driven by sequential data transformation workflows. These workflows begin with external dataset ingestion and progress through preprocessing, feature extraction, machine learning inference, and decision-support processes, ultimately culminating in alert generation and user presentation.

The decomposition is represented using:

- A Context-Level Data Flow Diagram (Level 0 DFD)
- A Level 1 Data Flow Diagram
- A Structural Decomposition Diagram

These representations collectively illustrate system boundaries, internal processing logic, data storage responsibilities, and the relationship between functional processes and architectural components.

3.2.1 Context-Level Data Flow Diagram (Level 0)

At the context level, BSADS is represented as a single processing entity (Process 0) interacting with external entities, as introduced in *Section 2.2*. This view defines the system boundary and highlights all external interactions.

The external entities include:

a) External Hive Monitoring Storage Platform
b) Mobile Application User (Beekeeper)
c) Weather Service APIs

The External Hive Monitoring Storage Platform provides pre-collected acoustic datasets along with associated metadata such as hive identifiers and timestamps. The Weather Service APIs supply environmental data, including temperature, humidity, and other meteorological parameters that enhance contextual analysis.

The Mobile Application User (Beekeeper) interacts with the system to access hive status information, receive alerts, and review historical reports. The user may also provide inputs such as configuration settings or alert acknowledgments.

Within this context, BSADS functions as a centralized analytical system that ingests external datasets, processes them, and returns actionable insights to the user.

3.2.2 Level 1 Data Flow Diagram

The Level 1 Data Flow Diagram decomposes Process 0 into major internal processes that correspond directly to the functional modules defined in Section 2.3 and the subsystems described in *Section 3.1*. This ensures consistency between functional behavior and architectural structure.

a) Major Processes

Process 1.0: Data ingestion and validation: This process receives datasets from external hive monitoring storage platforms and, where applicable, weather service APIs. It validates data format, checks metadata completeness, assigns unique identifiers, and ensures that only well-structured data enters the system.

Process 2.0: Data preprocessing and feature extraction: This process transforms validated acoustic datasets into structured representations suitable for machine learning. It performs signal normalization, segmentation, noise reduction, and extraction of relevant acoustic features such as frequency components and spectral characteristics.

Process 3.0: Machine learning inference: This process applies pre-trained machine learning models to the extracted feature vectors in order to classify hive conditions. The output consists of hive-state classifications (e.g., Normal, Pre-swarm, Swarm, Abscondment, or Uncertain) along with associated confidence scores.

Process 4.0: Alert and result management: This process evaluates classification outputs and determines whether alert conditions are met. It generates alerts where necessary, updates historical records, and prepares structured results for downstream consumption.

Process 5.0: Advisory and decision support: This process provides actionable recommendations based on the classified hive state and historical behavioral patterns. It translates analytical results into practical guidance to support beekeeper decision-making.

Process 6.0: Data storage and retrieval management This process manages interaction with system data stores. It ensures that raw datasets, extracted features, inference results, and alert records are properly stored and can be efficiently retrieved for processing, reporting, and historical analysis.

b) Data Stores

D1: Raw dataset repository

Stores validated acoustic metadata and environmental parameters received from external systems.

D2: Analytical results repository

Stores extracted feature vectors, machine learning outputs, hive status history, and generated alert records.

c) Logical Data Flow

The overall data flow within the system follows a structured and sequential pipeline. External hive monitoring platforms and weather APIs provide acoustic and environmental datasets to Process 1.0, where the data is validated and stored in the Raw Dataset Repository (D1).

Process 2.0 retrieves validated data from D1 and performs preprocessing and feature extraction. The resulting feature vectors are then passed to Process 3.0, where machine learning inference is performed to classify hive states.

The classification results are stored in the Analytical Results Repository (D2).

Process 4.0 retrieves these results, evaluates alert conditions, and prepares structured outputs. Process 5.0 further enriches these outputs by generating advisory recommendations.

Finally, the processed results, alerts, and advisory information are made available to the mobile application through backend APIs, enabling user interaction and decision-making.

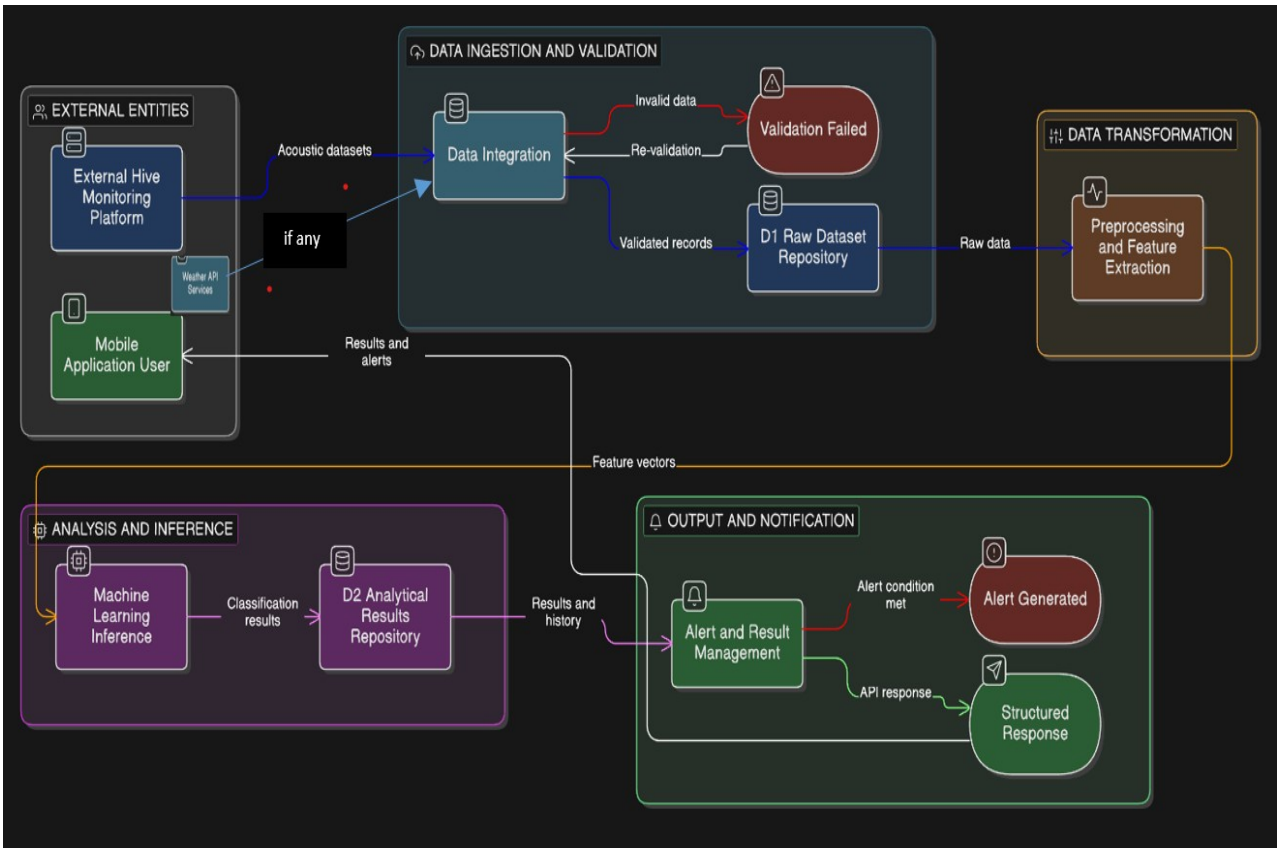

Figure 3. 2 Level 1 Data Flow Diagram for BSADS

Figure 3.2 illustrates the Level 1 Data Flow Diagram's visual representation of data flow within the system as processes execute and the corresponding subsystems

3.2.3 Structural Decomposition Diagram

While the data flow diagram above represents data movement, the structural decomposition diagram below illustrates the hierarchical organization of the sub system modules.

The BSADS can be structurally decomposed as follows:

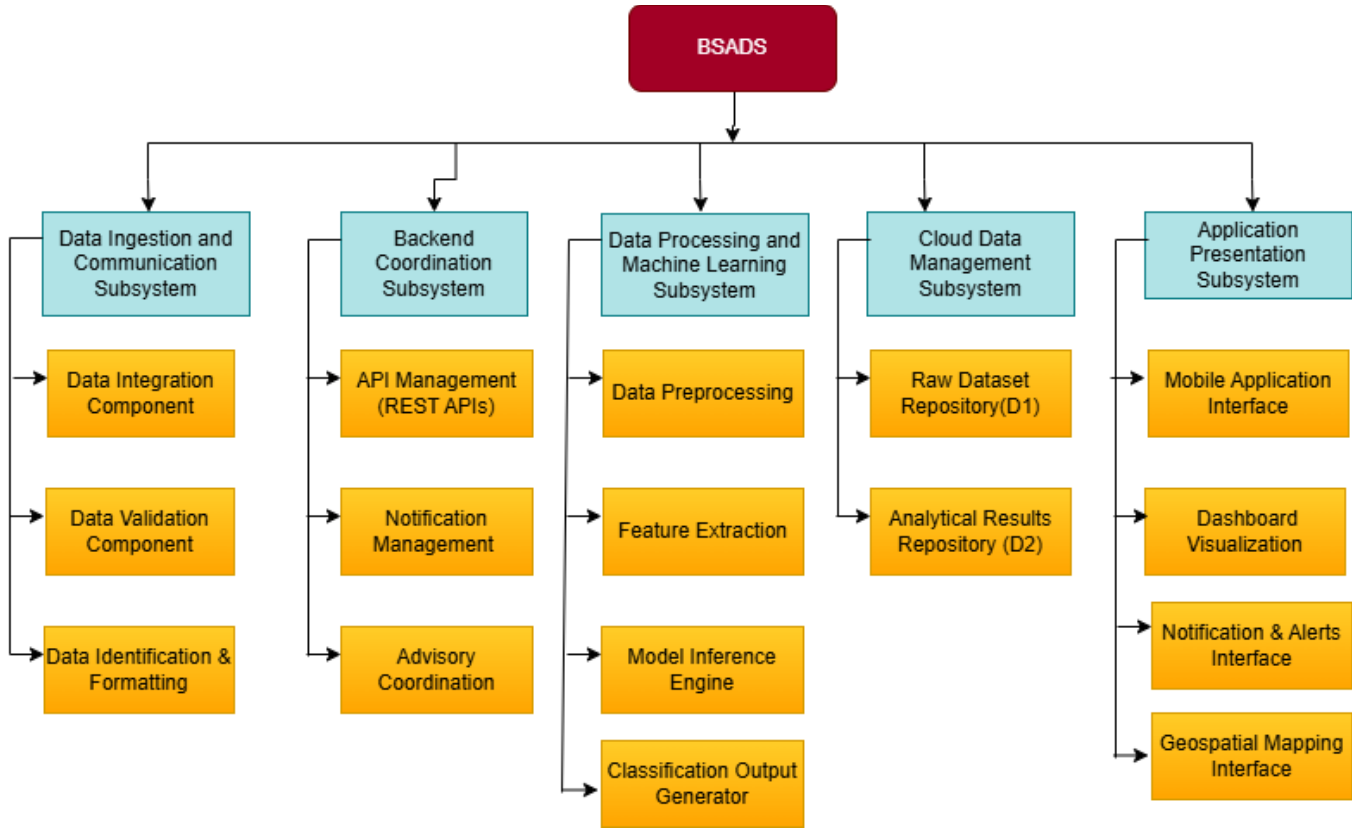

Figure 3. 3 Structural Decomposition Diagram for BSADS

Figure 3.3 illustrates the structural decomposition of the BSADS. The diagram presents a hierarchical breakdown of the system into its major subsystems and their internal components. Each subsystem corresponds to a specific responsibility within the overall system architecture, including data integration, processing, coordination, storage, and presentation. This decomposition ensures clear separation of concerns and demonstrates how high-level system functionality is distributed across structurally independent but interconnected components.

3.2.4 Data Flow Integrity and Consistency

The decomposition ensures a consistent, traceable pipeline:

- External datasets are validated before processing.
- All preprocessing occurs before machine learning inference.
- Classification results are persisted before alert evaluation.
- The mobile application accesses data only through authenticated backend interfaces.

No component bypasses another layer, thereby preserving data integrity and architectural separation of concerns.

3.3 Design Rationale

The modular layered architecture described in *Section 3.1* was selected to address the functional, technical, and operational requirements of the Bee Swarming and Abscondment Detection System.

3.3.1 Rationale for Selected Architecture

The system adopts a layered, modular architecture to achieve the following objectives:

i) **Scalability:** New sensors, hives, or machine learning models can be added without major changes to existing components.
ii) **Maintainability:** Each subsystem can be updated or replaced independently, reducing system downtime and development complexity.
iii) **Fault Isolation:** Failures in one subsystem (e.g., communication issues) do not propagate to other subsystems.
iv) **Flexibility:** The architecture supports multiple communication technologies and future system extensions.

Separating data acquisition, processing, and presentation allows the system to handle real-time monitoring while supporting long-term data analysis.

3.3.2 Trade-offs Considered

Several trade-offs were evaluated during architectural selection:

- **Cloud-based processing vs. Edge processing:**
  Cloud-based processing was chosen over full edge-based analytics to reduce computational burden on resource-constrained microcontrollers. This improves model accuracy and simplifies updates, at the cost of dependency on network connectivity.
- **Modular architecture vs. Monolithic architecture:**

A monolithic design was considered but rejected due to poor scalability and difficulty in maintenance. The modular approach improves system evolution but introduces slightly higher integration complexity.

- **Real-time processing vs. Energy efficiency**:
  Continuous real-time monitoring increases energy consumption. The design balances this by using scheduled sensing intervals and lightweight preprocessing on the hive nodes.

3.3.3 Alternative Architectures Considered

A fully edge-based architecture, where all data processing and machine learning inference occur on the hive node, was considered. However, this approach was not selected due to limited processing power, memory constraints, and increased energy consumption on embedded devices.

A centralized monolithic server architecture was also evaluated but rejected due to limited scalability and reduced fault tolerance.

3.3.4 Conclusion

The selected modular layered architecture provides an optimal balance between performance, scalability, maintainability, and resource efficiency. It supports reliable hive monitoring, accurate predictive analysis, and effective user interaction while allowing future enhancements and integration with broader smart agriculture systems.

4. DATA DESIGN

4.1. Data Description

The data design of the Bee Swarming and Abscondment Detection System (BSADS) defines the structure, organization, and lifecycle of data as it flows through the system. The system operates on pre-collected acoustic datasets retrieved from external storage platforms. As such, BSADS does not control upstream data acquisition but instead focuses on transforming externally stored datasets into meaningful analytical outputs.

Due to the large size and bandwidth requirements of raw audio files, the system does not store audio data directly within the database. Instead, it maintains references (Uniform Resource Identifiers - URIs) to externally stored audio files. This design improves scalability, reduces storage overhead, and enables efficient integration with external storage systems.

The data architecture is organized into three major layers:

i) Data Ingestion and Preprocessing Layer
ii) Machine Learning Processing Layer
iii) Application and Persistent Storage Layer

These layers collectively ensure data integrity, traceability, scalability, and secure access across backend services and the mobile application.

Additionally, the system distinguishes between:

- Persistent data entities stored in relational databases
- Generated analytical data such as feature vectors, inference results, and processed visual artifacts (such as spectrogram images).

4.2 Data Organization by System Component

4.2.1 Data ingestion & preprocessing

The Data Sanitization Component manages raw datasets retrieved from cloud storage. It validates, cleans, and prepares acoustic and environmental data before feature extraction.

Table 4. 1 Raw Acoustic Data Entity

Field Name	Data Type	Description
audio_id	UUID	Unique identifier for the audio record
hive_id	Integer	Associated hive
source_uri	Varchar(255)	External storage location (file path or URL)
file_format	Varchar(20)	Audio format (e.g., WAV, MP3)
duration_seconds	Decimal(6,2)	Length of recording
captured_at	DateTime	Time of recording
ingestion_timestamp	DateTime	Time data was ingested

Table 4.1 The audio source entity represents externally stored acoustic recordings collected from beehives. Instead of storing raw audio files within the system, this entity maintains references (URIs) to their storage locations. This design minimizes storage overhead and improves scalability. Each audio record is associated with a specific hive and includes metadata such as format, duration, and capture time. These recordings form the primary input to the signal processing and machine learning pipeline.

Table 4. 2 Raw Environmental Data Entity (Optional)

Field Name	**Data Type**	**Description**
env_record_id	UUID	Unique identifier
hive_id	Integer	Associated hive
temperature	Decimal(5,2)	Temperature in °C
humidity	Decimal(5,2)	Relative humidity (%)
recorded_at	DateTime	Time of measurement

Table 4.2 Environmental parameters provide contextual information that may improve model accuracy. These records are optional and may not be available for all datasets.

Table 4. 3 Sanitized Audio Metadata Entity (Transient)

Field Name	Data Type	Description
sanitized_id	UUID	Unique identifier
audio_id	UUID	Reference to audio source
noise_reduced	Boolean	Indicates noise filtering
normalized	Boolean	Indicates normalization
segmentation_window_ms	Integer	Segmentation window size

Table 4.3 This entity represents intermediate preprocessing metadata. It is not permanently stored but may be used temporarily for traceability, debugging, and reproducibility during feature extraction..

4.2.2 Machine Learning Processing Component

This component transforms processed acoustic signals into structured representations and generates classification outputs.

Table 4. 4 Feature Vector Entity

Field Name	Data Type	Description
feature_id	UUID	Unique identifier
audio_id	UUID	Source audio reference
hive_id	Integer	Associated hive
mfcc_values	Text	MFCC coefficients
spectral_centroid	Float	Frequency centroid
spectral_bandwidth	Float	Spectral spread
signal_energy	Float	Signal energy
zero_crossing_rate	Float	Signal variation

temperature_normalized	Float	Normalized temperature
humidity_normalized	Float	Normalized humidity
created_at	DateTime	Feature extraction time

Table 4.4 defines the numerical representation of acoustic signals used for machine-learning inference. MFCC coefficients and spectral features form the core predictors of hive state classification. Environmental features are normalized to maintain scale consistency. Feature vectors represent numerical abstractions of acoustic signals used for machine learning inference. Although features can be generated dynamically, they may be stored to support traceability, reproducibility, and future analytical task

Table 4. 5 Model Metadata Entity(optional)

Field Name	Data Type	Description
model_id	UUID	Unique model identifier
model_type	Varchar(50)	Algorithm type
training_dataset_size	Integer	Number of training samples
training_accuracy	Decimal(5,2)	Training accuracy (%)
validation_accuracy	Decimal(5,2)	Validation accuracy (%)
model_version	Varchar(20)	Version label
deployment_date	DateTime	Deployment timestamp

Table 4.5 stores information about trained models. Maintaining model metadata supports reproducibility, version control, and performance evaluation.

Table 4. 6 Inference Result Entity

Field Name	Data Type	Description
inference_id	UUID	Unique identifier
feature_vector_id	UUID	Source feature vector
hive_id	Integer	Associated hive
hive_state	Varchar(30)	Classified hive condition

confidence_score	Decimal(5,4)	Prediction probability (0–1)
inference_latency_ms	Integer	Processing time
timestamp	DateTime	Time of inference

Table 4.6 captures classification outputs generated by the deployed machine learning model. It forms the basis for alert generation and advisory recommendations.

Table 4. 7 Processed Image Entity

Field Name	Data Type	Description
image_id	UUID	Unique identifier
audio_id	UUID	Associated audio
hive_id	Integer	Associated hive
image_type	Varchar(50)	Type (e.g., spectrogram)
image_uri	Varchar(255)	Storage path
generated_at	DateTime	Creation time

Table 4.7 stores references to visual artifacts generated during signal processing, such as spectrograms. These images support visualization, debugging, and interpretability of model behavior.

4.2.3 Application and Persistent Storage Component

This layer supports user management, hive tracking, alerting, and historical analytics in addition to decision support.

Table 4. 8 User Entity

Field Name	Data Type	Description
user_id	Integer	Primary key
full_name	Varchar(30)	User name
telephone_number	Varchar(15)	Contact number

email	Varchar(50)	Unique login email
password_hash	Varchar(255)	Hashed password
role	Varchar(30)	User role
created_at	DateTime	Account creation date

Table 4.7 The user entity represents individuals who interact with the BSADS platform. These include beekeepers and system stakeholders responsible for monitoring hive conditions and responding to alerts. The entity stores essential authentication and identification information such as user credentials, contact details, and system roles.

Table 4. 9 Hive Entity

Field Name	Data Type	Description
hive_id	Integer	Primary key
owner_id	Integer	Foreign key to User
hive_location	Varchar(150)	Hive location
hive_type	Varchar(50)	Hive model
installation_date	Date	Registration date
current_state	Varchar(30)	Latest hive state

Each hive belongs to a single user but may generate multiple sensor audio data and inference records.

Table 4. 10 Alert Entity

Field Name	Data Type	Description
alert_id	UUID	Primary key
inference_id	UUID	Triggering inference
hive_id	Integer	Associated hive
alert_type	Varchar(30)	Swarm / Abscondment
severity_level	Varchar(20)	Alert severity
status	Varchar(20)	Pending / Acknowledged
generated_at	DateTime	Alert time

Table 4.10 The Alert entity represents notifications generated when abnormal hive conditions are detected. Alerts are triggered based on inference results and categorized by type and severity level. This entity enables timely communication with the beekeeper, allowing for rapid response to potential risks such as swarming or abscondment. The alert status field supports tracking whether an alert has been acknowledged or acted upon.

Table 4. 11 Advisory Entity

Field Name	Data Type	Description
advisory_id	UUID	Primary key
inference_id	UUID	Associated inference
hive_id	Integer	Associated hive
advisory_type	Varchar(30)	Preventive / Reactive
created_at	DateTime	Creation time

Table 4.11 The **Advisory** entity represents a structured decision-support instance generated in response to a specific inference result. It groups a set of recommended actions intended to guide the beekeeper in addressing detected hive conditions. Rather than providing a single recommendation, the advisory serves as a container for multiple actionable steps, ensuring that guidance is organized, comprehensive, and context-specific. This design supports systematic intervention and improves the usability of the system's decision-support functionality.

Table 4. 12 Advisory Action Entity (Checklist)

Field Name	Data Type	Description
action_id	UUID	Primary key
advisory_id	UUID	FK to Advisory
action_description	Text	Recommended action
priority_level	Varchar(20)	Priority
status	Varchar(20)	Pending / Completed
created_at	DateTime	Timestamp

Table 4.12 The **Advisory Action** entity defines individual recommended actions associated with an advisory instance. Each advisory may contain multiple actions, forming a checklist that the beekeeper can follow. This structure allows actions to be prioritized and tracked independently, enabling users to monitor progress and ensure that all necessary interventions are completed. It enhances usability by transforming analytical outputs into clear, actionable steps

4.3 Entity Relationship Structure

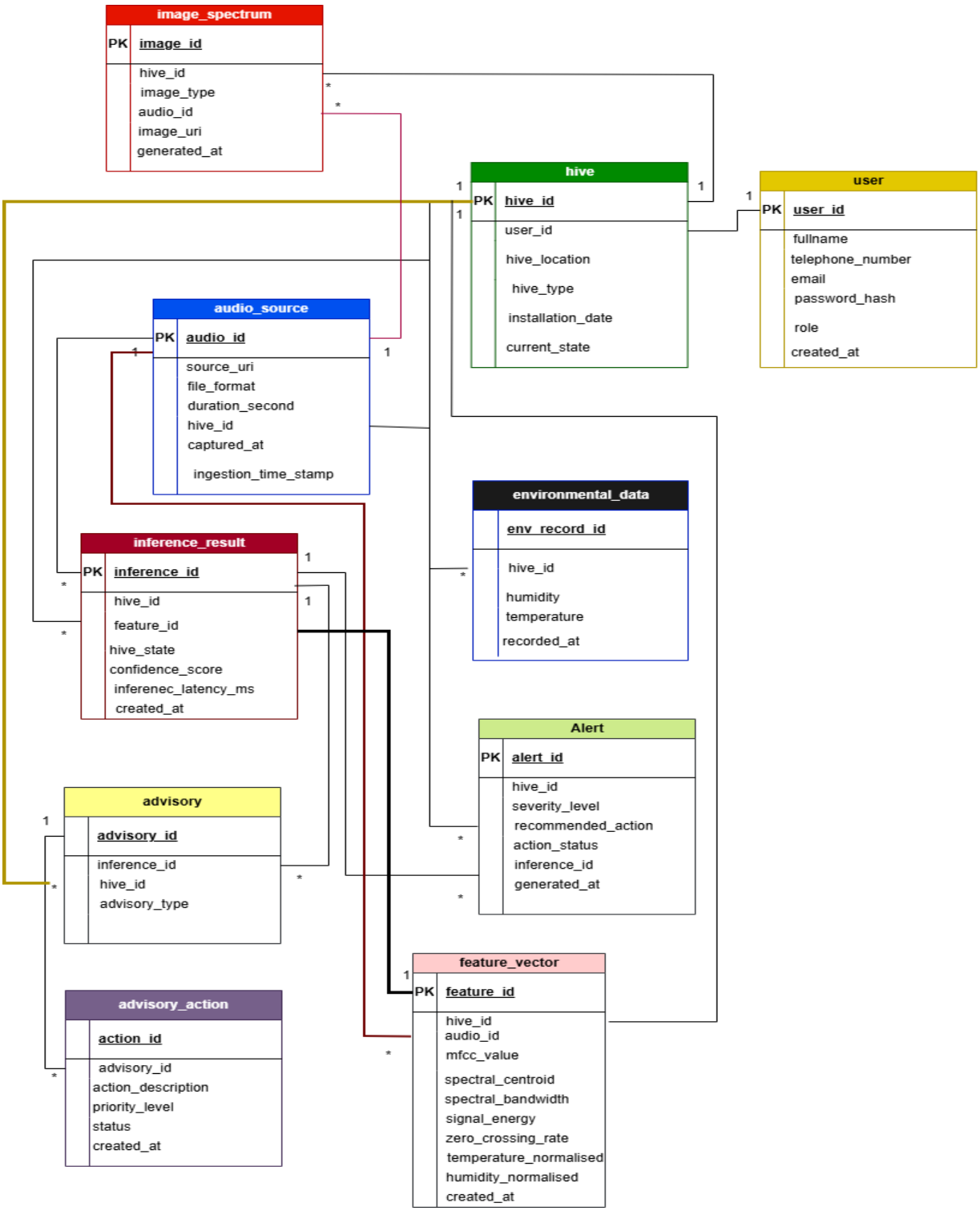

Figure 4. 1 illustrates the logical relationships among core persistent entities.

Each hive generates multiple feature vectors and inference results over time. Alerts are triggered by specific inference records. This relational structure ensures traceability from raw datasets to user-facing notifications.

Transient entities such as sanitized audio metadata are excluded from the ERD since they are not permanently stored.

4.4 Data Integrity and Storage Considerations

The system enforces referential integrity through foreign key constraints between User, Hive, Inference, and Alert entities. Feature vectors may be stored either in structured relational tables or serialized formats depending on storage optimization strategies.

5. COMPONENT DESIGN

The BSADS system adopts a modular and service-oriented design approach in which each component performs a well-defined function within the overall machine-learning processing pipeline. The component design is derived directly from the system architecture described in *Chapter 3* and ensures traceability to the functional requirements specified in the Software Requirements Specification (SRS).

The BSADS system is implemented primarily as a backend-driven architecture with supporting mobile client components.

5.1 Component Overview

The major components of the BSADS system include:

1. Data Integration Component
2. Preprocessing and Feature Extraction Component
3. Machine Learning Inference Component
4. Cloud Synchronization Component
5. System Monitoring and Logging Component

Each component is described in detail in the following sections.

5.1.1 Data Integration Component

Purpose

Acts as the primary ingestion layer for hive-related data. As BSADS does not perform direct sensor acquisition, this component serves as the central gateway for all data transmitted from external monitoring systems.

Inputs

- Pre-recorded hive audio files
- Environmental parameters (temperature, humidity)
- Metadata (hive ID, timestamp, location)

Outputs

- Validated and structured raw data records
- Trigger signal for preprocessing pipeline

Responsibilities

- Accept data streams from storage systems through secure REST endpoints
- Validate file formats and metadata completeness
- Assign unique identifiers to each dataset
- Forward validated data for storage and preprocessing

Processing Logic

1. *Receive incoming data request*
2. *Validate authentication token*
3. *Check audio file format and size*
4. *Validate metadata fields*
5. ***If** validation fails:*

    Reject the request and log an error

    Else:

    Generate a unique data ID

    Store raw data in a repository

    Trigger preprocessing pipeline

Figure 5. 1 Process Logic for Data Integration Component

5.1.2 Preprocessing and Feature Extraction Component

Purpose

This component prepares raw hive audio data for machine-learning analysis by cleaning, segmenting, and transforming signals into numerical feature representations.

Inputs

- Raw audio recordings
- Environmental readings

Outputs

- Acoustic feature vectors
- Normalized environmental feature vectors

Responsibilities

- Perform noise reduction
- Normalize signals

- Segment audio windows
- Extract acoustic features (such as MFCCs, spectral features)
- Merge environmental parameters

Processing Logic

```
1. Load raw audio file
2. Apply a noise reduction filter
3. Normalize amplitude levels
4. Segment audio into fixed windows
5. For each segment:
      Extract MFCC and spectral features
6. Normalize environmental parameters
7. Merge acoustic and environmental features
8. Store feature vectors in the database
```

Figure 5. 2 Processing Logic for 2 Preprocessing and Feature Extraction Component

5.1.3 Machine Learning Inference Component

Purpose

Manages persistent storage of system data and supports historical analysis.

Inputs

- Raw metadata
- Feature vectors
- Classification results

Outputs

- Structured and indexed database records

Responsibilities

- Maintain relational database integrity
- Store metadata and results
- Support historical queries

- Enable dataset retrieval for retraining

Processing Logic

1. *Load trained ML model into memory*
2. *Receive feature vector batch*
3. *Apply feature normalization*
4. *Perform inference using the model*
5. *Compute prediction probabilities*
6. *Assign hive state label*
7. Return classification results

Figure 5. 3 Processing Logic for Machine Learning Inference Component

5.1.4 Cloud Synchronization Component

Purpose

Ensures long-term archival, redundancy, and cross-service availability of processed datasets by synchronizing classification results and analytical summaries with cloud-based storage services.

Inputs

- Classification results
- Analytical summaries
- Processed metadata records

Outputs

- Archived cloud datasets
- Synchronization status logs

Responsibilities

- Perform scheduled synchronization of processed data
- Compress datasets before upload

- Encrypt archived data packages
- Tag synchronized datasets with metadata
- Maintain redundancy and backup copies

Processing Logic

At scheduled intervals:

1. *Retrieve new classification records*
2. *Compress data files*
3. *Upload to cloud storage*
4. *Update synchronization status*
5. *Log synchronization success or failure*

Figure 5. 4 Processing Logic for Machine Learning Inference Component

5.1.5 System Monitoring and Logging Component

Purpose

Provides operational transparency, reliability assurance, and diagnostic support by continuously monitoring system activities and recording significant events across all components.

Inputs

- Data ingestion events
- Preprocessing and inference events
- API access requests
- Synchronization status updates
- System performance metrics

Outputs

- Structured log records
- Error reports

- Performance summaries

Responsibilities

- Capture application-level events
- Record processing errors and exceptions
- Monitor model inference execution
- Track API authentication attempts
- Generate alerts for critical failures

Processing Logic

1. *Capture event from system component*

2. *Classify event type (info, warning, error, critical)*
3. *Record timestamp and source component*
4. *Store log entry in monitoring repository*
5. *Trigger alert notification if the severity threshold is exceeded*

Figure 5. 5 Processing Logic for System Monitoring and Logging Component

6. HUMAN INTERFACE DESIGN

6.1 Overview of User Interface

BSADS provides a mobile application–based user interface through which beekeepers access hive analytics, predictive alerts, and historical behavioral insights. The interface is designed to prioritize usability, clarity, and responsiveness, enabling users with limited technical expertise to interpret system outputs and make informed management decisions.

Upon launching the application, users are presented with a centralized dashboard summarizing the current status of monitored hives. This summary is derived from processed acoustic and environmental data received from integrated hive-monitoring systems. The dashboard displays the predicted hive conditions, such as Normal, Pre-swarm, Swarm, Abscondment, Pest Infestation, or Uncertain, together with key environmental indicators and intuitive visual elements, including status labels and graphical representations.

The application allows users to navigate to detailed analytical views, where historical trends of hive states and associated environmental parameters can be examined over time. This functionality supports pattern recognition and long-term assessment of hive behavior.

In addition, the interface delivers alert notifications when analytical results indicate elevated risks, such as imminent swarming or abscondment. These alerts enable proactive intervention by beekeepers, reducing reliance on frequent manual inspections and improving response timeliness.

Overall, the BSADS user interface emphasizes effective communication of complex analytical results through a simplified and visually structured presentation, supporting practical, data-driven hive management.

6.2 Screen Images

This section presents the proposed user interface screens of the BSADS mobile application. The screens illustrate the complete user interaction flow, from authentication and dashboard access to

hive analytics and alert monitoring. The interfaces were designed as mobile application mockups and represent the intended final appearance of the system.

6.2.1 Authentication Screens

i) Splash Screen

Displays the BSADS logo and tagline, providing an entry point to the application.

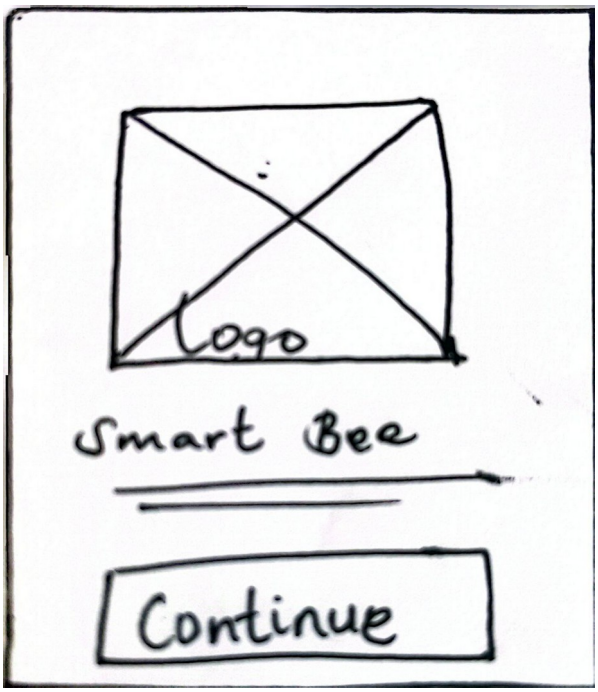

Figure 6. 1 Landing screen for the mobile application

ii) Login Screen

This allows registered users to authenticate using their credentials.

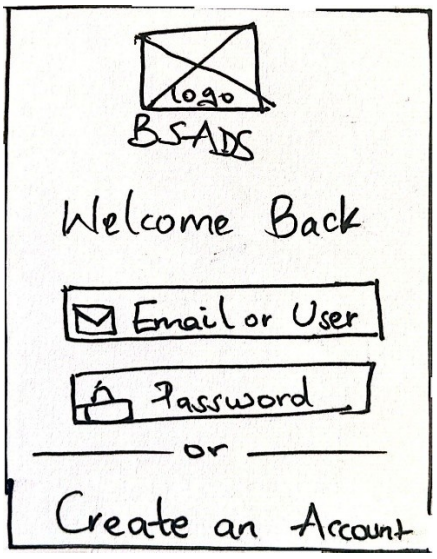

Figure 6. 2 Login screen for the users to access the system

ii) Sign-Up Screen

Enables new users to create an account by providing the required registration details.

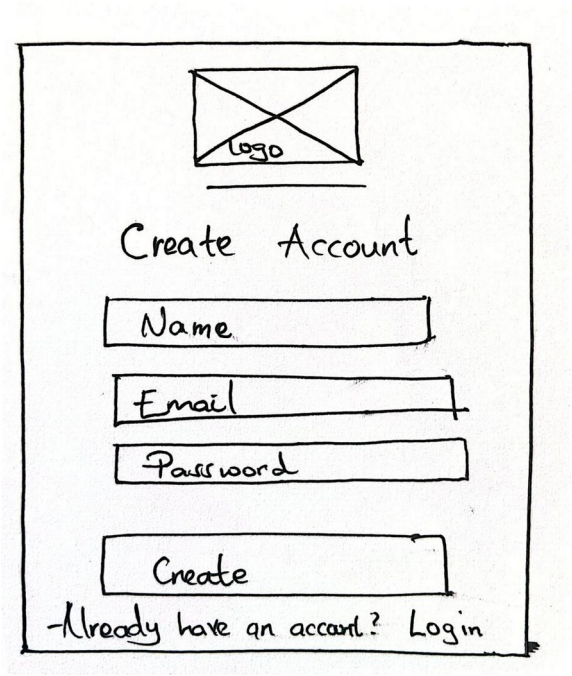

Figure 6. 3 Signup screen for new users to register accounts

6.2.2 Core Monitoring Screens

i) Dashboard Screen

Provides a high-level overview of hive health status, key metrics shown on the left, and on the right, recent alerts.

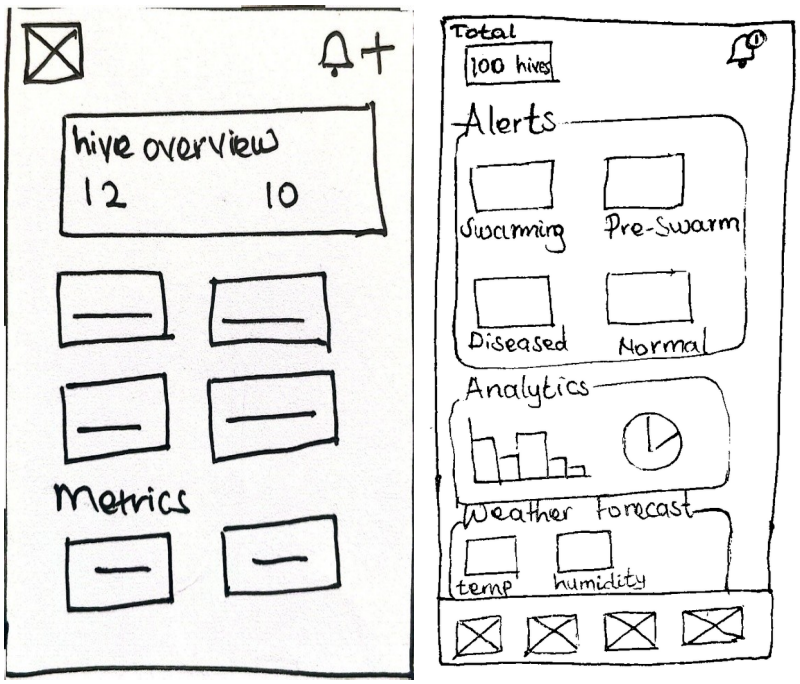

Figure 6. 4 Hive monitoring screen(left) and Dashboard(right)

ii) All Hives Screen

Displays a list of all hives associated with the user, showing their current status and enabling quick navigation to individual hive details.

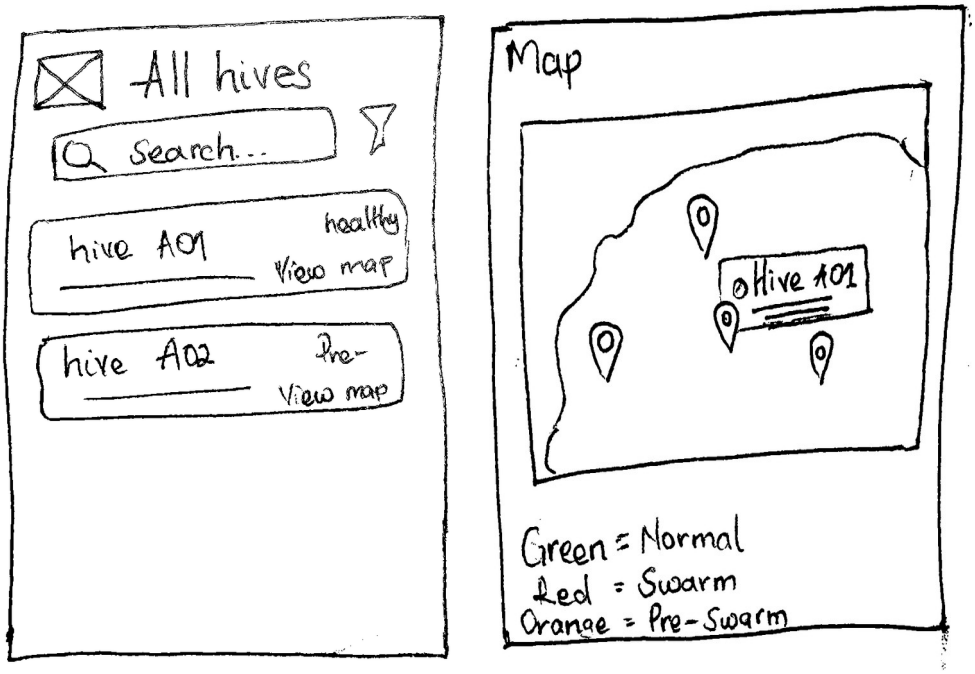

Figure 6. 5 All hives monitoring screen to visualize all hive status(left) and map(right).

iii) Hive Details Screen

Presents detailed analytical information for a selected hive, including predicted hive condition, historical environmental trends, and acoustic activity patterns.

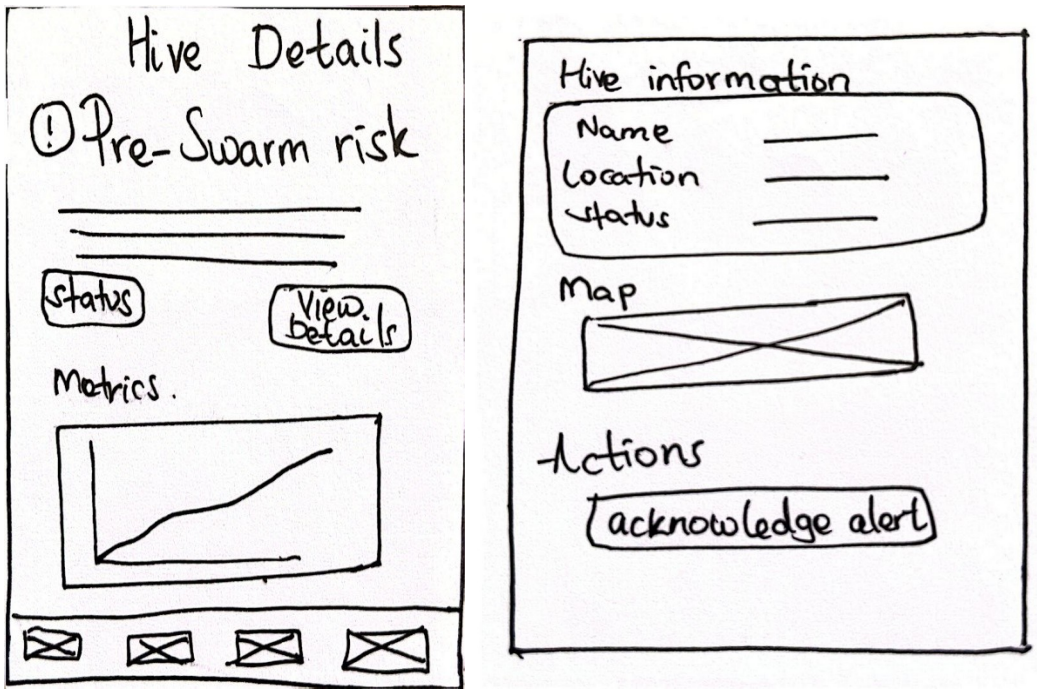

Figure 6. 6 Hive Details screen and information(left) and More hive Details visualizations(right)

6.2.3 Alerts and Notifications Screens

i) Alerts

Displays a list of system-generated alerts related to potential swarming or abscondment risks.

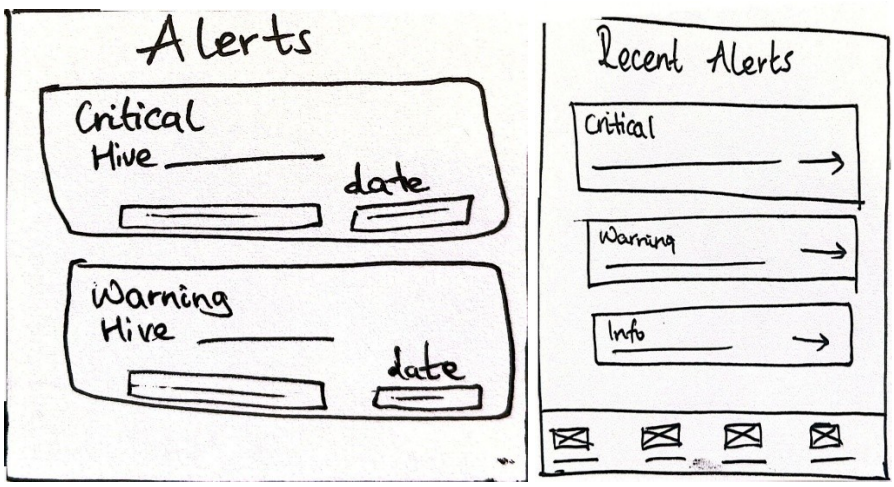

Figure 6. 7 Alerts Summary screen

ii) Alert Details Screen

Provides in-depth information about a selected alert, including risk level, contributing factors, and recommended actions.

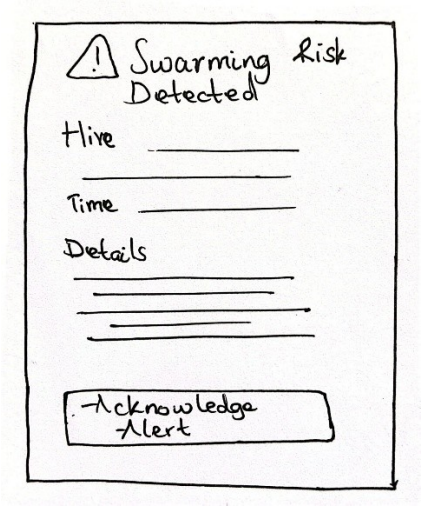

Figure 6. 8 Alert Details screen

6.3 Screen Objects and Actions

These are the main screen objects used within the BSADS mobile application and the actions associated with user interaction on each screen. Screen objects include input fields, buttons, indicators, charts, and navigation elements. These objects enable users to interact with the system and receive feedback from the underlying analytics and machine learning processes.

a) Splash Screen

The splash screen contains the application logo and system tagline.
Action: The screen automatically transitions to the login screen after initialization.

b) Login Screen

This screen contains text input fields for email and password, as well as a login button. **Actions:**

- User credentials are submitted for authentication.

- Error messages are displayed for invalid input or failed login attempts.

c) Sign-Up Screen

The sign-up screen includes user registration form fields and a create account button.
Actions:

- User details are validated and stored.
- Successful registration redirects the user to the login screen.

d) Dashboard Screen

The dashboard displays summary metric cards, hive status indicators, and recent alert notifications.
Actions:

- Presents real-time system analytics.
- Automatically updates when new data or predictions are available.

e) All Hives Screen

This screen displays a list of all monitored hives with status labels.
Actions:

- Selecting a hive navigates the user to the hive details screen.

f) Hive Details Screen

The hive details screen includes predictive status indicators, historical charts (temperature, humidity, and acoustic activity), and hive metadata.
Actions:

- Allows users to analyze historical trends.
- Provides insight into predicted swarming or abscondment risk.

g) Alerts Screens

These screens display alert lists, severity indicators, and alert details.

Actions:

- Notifies users of detected risks.
- Allows users to view detailed alert information and acknowledge alerts.

h) Settings Screen

The settings screen contains profile information fields, notification toggles, and system configuration options.

Actions:

- Allows customization of notification preferences and user profile management.

7. REQUIREMENTS MATRIX

This section provides a requirements traceability matrix that cross-references the **functional requirements defined in the Software Requirements Specification (SRS)** with the **system components.** The matrix ensures that every functional requirement is addressed by one or more system components.

7.1 Functional Requirements to Components Traceability Matrix

✔ = Component satisfies the requirement

Table 7. 1 Requirements Matrix

SRS Requirement ID	Functional Requirement Description	Data Integration Component	Preprocessing & Feature Extraction Component	Machine Learning Analysis Component	Backend & Data Management Component	Mobile Application Component	Database / Storage Component
FR-1.1	Integrate acoustic hive sound data	✔					✔
FR-1.2	Integrate environmental data (temperature, humidity)	✔					✔
FR-1.3	Handle missing or inconsistent data	✔	✔				✔
FR-2.1	Preprocess acoustic		✔				

	and environmental data						
FR-2.2	Extract relevant features from input data		✔				
FR-2.3	Classify hive states using machine learning			✔			✔
FR-2.4	Store classification results				✔		✔
FR-3.1	Generate alerts for swarming risk			✔	✔	✔	
FR-3.2	Generate alerts for abscondment risk			✔	✔	✔	
FR-4.1	Display current hive status				✔	✔	
FR-4.2	Display historical hive data and trends				✔	✔	✔
FR-4.3	Allow users to view alert details				✔	✔	
FR-4.4	Support user authentication				✔	✔	

7.2 Summary

The Requirements Matrix demonstrates that all functional requirements defined in the SRS are fully supported by the system components. Each requirement is mapped to one or more components, ensuring consistency between system requirements and system design.